

PRIMARY ISSUE CATEGORY: TORQUE SPIKE

1. PROBLEM DEFINITION

A torque spike is a rapid, significant increase in surface (or downhole) torque, often driven by bit whirl, stick-slip vibration, hard/abrasive stringers, or elevated side loads at dogleg points, and can precede bit or BHA damage if not addressed promptly.

2. EARLY WARNING SIGNS

- Torque rising sharply (e.g. >1.5x baseline) over a short interval (minutes).
- Erratic RPM at surface despite a constant top-drive/rotary setting (stick-slip signature).
- Downhole vibration data (if available) showing lateral or torsional vibration spikes.
- Entering a known hard/abrasive stringer or a section with elevated dogleg severity.

3. ROOT CAUSES (FIELD_ALPHA HISTORY)

- Hard, abrasive calcareous (or similar) stringers causing bit whirl.
- Concentrated dogleg severity from a directional build elevating side loads on the BHA.
- Friction factor mismatch between the pre-well torque-and-drag model and actual hole conditions.
- Insufficient lubricant/friction-reducing additive in the active mud system.

4. STEP-BY-STEP SOLUTION / RESPONSE PROCEDURE

Step 1: Reduce WOB and RPM — Cut weight-on-bit by roughly 25-30% and reduce RPM (typically to 65-75) to break a stick-slip cycle.

Step 2: Compare torque-and-drag model — Rerun the real-time torque-and-drag model against actual surface torque to identify whether the friction factor assumption is too low for the interval.

Step 3: Perform a short wiper trip — Trip back through the build/tangent section to clean the hole and confirm no mechanical restriction remains.

Step 4: Add lubricant additive — Add an ester-based (or equivalent) lubricant to the active mud system, typically at ~2% concentration, to reduce friction.

Step 5: Resume at reduced ROP — Resume drilling at a reduced ROP through the remaining stringer/dogleg interval as a precaution, monitoring torque continuously.

5. MATERIALS / PARAMETERS QUICK REFERENCE

Response	Typical Adjustment
WOB reduction	25-30% below pre-spike value
RPM reduction	To 65-75 RPM to break stick-slip
Lubricant additive	2% ester-based lubricant, maintained through the interval
Updated friction factor	Increase model assumption (e.g. 0.25 -> 0.32) for the specific stringer/dogleg zone

6. PREVENTION CHECKLIST FOR OFFSET WELLS

- Add lubricant additive proactively before entering a known hard-stringer/dogleg interval, rather than reacting after a spike.
- Specify an anti-whirl bit design for repeat offenders identified via dull grading history.

- Coordinate with directional drilling to keep dogleg severity below ~3.5 deg/30m through problem intervals where feasible.
- Run downhole vibration monitoring tools in intervals with a history of torque instability.

7. FIELD CASE REFERENCES

Well	Depth	Outcome
W005	2740 m	Torque 10->22 kft-lb resolved via WOB/RPM reduction + wiper trip + 2% lubricant

This Issue Resolution Playbook is a synthetically generated reference document produced for the NWIS prototype, consolidating lessons across Field_Alpha and Field_Alpha-adjacent wells. Not associated with an actual drilling operation.